

[6]

Petrology and chemistry of two “large” granite clasts from the Moon

Paul H. Warren¹, G. Jeffrey Taylor¹, Klaus Keil¹, David N. Shirley² and John T. Wasson^{2,*}

¹ Department of Geology and Institute of Meteoritics, University of New Mexico, Albuquerque, NM 87131 (U.S.A.)

² Institute of Geophysics and Planetary Physics, Department of Earth and Space Sciences, University of California, Los Angeles, CA 90024 (U.S.A.)

Received December 21, 1982
Revised version received April 11, 1983

Pristine granite clasts in Apollo-14 breccias 14321 and 14303 have estimated masses of 1.8 and 0.17 g, respectively. The 14321 clast is ~ 60% K-feldspar and 40% quartz, with traces of extremely Mg-poor mafic silicates and ilmenite. The 14303 clast is roughly 33% plagioclase, 32% K-feldspar, 23% quartz, 11% pyroxene, and 1% ilmenite; pyroxene and ilmenite are moderately Mg-rich; plagioclase and pyroxene are strongly zoned. Both clasts are severely brecciated, but monomict (pristine). Both have abundant graphic intergrowths of K-feldspar with quartz. Unlike the majority of similar Earth rocks, both clasts are devoid of hydrous phases. The bulk composition of the 14321 clast is similar to those of several other lunar granitic samples, but the 14303 clast is unique: it bears as close a resemblance to KREEP as it does to other lunar granites. Silicate liquid immiscibility may explain why the granites are low in REE relative to KREEP.

1. Introduction

Granite is abundant in Earth's crust, and before the Surveyor program, some (e.g. [1–4]) speculated that granitic rocks predominate in the lunar crust. But few granites turned up among lunar samples, and until now all were either obscure components in polymict breccias (e.g., 12013), or small clasts that could be unrepresentative pieces of diorites. Even the tiny mesostasis portions of fine-grained basaltic rocks were discussed under the heading of lunar “granites.” Valuable information came from such samples, but heretofore no large, pristine granites were known. In lunar jargon, a pristine rock is one that is compositionally endogenous (igneous or thermally metamorphosed igneous), unadulterated by other materials. The vast major-

ity of lunar highlands rocks are non-pristine, polymict products of impact-induced mixing of lunar plus minor meteoritic materials. The complex problem of how to identify pristine samples was reviewed by Warren and Wasson [5]. We have discovered and analyzed (for experimental procedures, see Warren et al. [6]) two pristine lunar granites, 10–100 times larger than any previously described. We also discuss the origin of this rare but important rock type.

2. Clast 14321,1027

The largest granite (14321,1027) was found as a 16 × 7 mm outcrop on breccia 14321 (total mass, 9 kg). NASA personnel estimated the original mass of the clast to be 1.8 g, of which about 1.5 g remains. Our chemical analysis consumed 149 mg. We also received (from two separate “chips”) three

* Also Department of Chemistry.

thin sections: 14321,993, 14321,1029, and 14321,1047.

In IUGS classification [7], this clast is an alkali-feldspar granite. Not counting the 30% that is shock-melted glass, the thin sections studied (combined area $\sim 30 \text{ mm}^2$) contain $\sim 60\%$ K-feldspar, 40% quartz, $< 1\%$ fayalite, and traces of ferrophedenbergite, ilmenite and Fe-Ni metal. The clast is highly brecciated, but in some areas an undamaged igneous texture is visible (Fig. 1). In such areas, feldspar and quartz generally occur as large intergrown crystals (graphic texture). Crystals are optically continuous over distances up to 1.8 mm, and the width of quartz blebs in feldspar averages $\sim 0.15 \text{ mm}$. These intergrowths are far

coarser than those of lunar granites described previously: 10–20 μm in a 0.02 g felsite clast from 73215 [8]; “on the order of a few microns” in two small clasts from 15405 [9]; roughly 5–40 μm in the numerous $< 2 \text{ mm}$ clasts from Boulder 1, Apollo 17 Station 2 (figure in Ryder et al. [10]; fig. III-14 in Stoesser et al. [11]); and 20–60 μm in several 1–2 mm clasts from 14305 [12]. The original texture of the only truly large granite, the one that is adulterated as a component of 12013, is totally obscured [13]. The 14321, 1027 texture is borderline between graphic sensu stricto, and “micrographic” (graphic, but invisible without a microscope). More specifically, it might be termed granophyric.

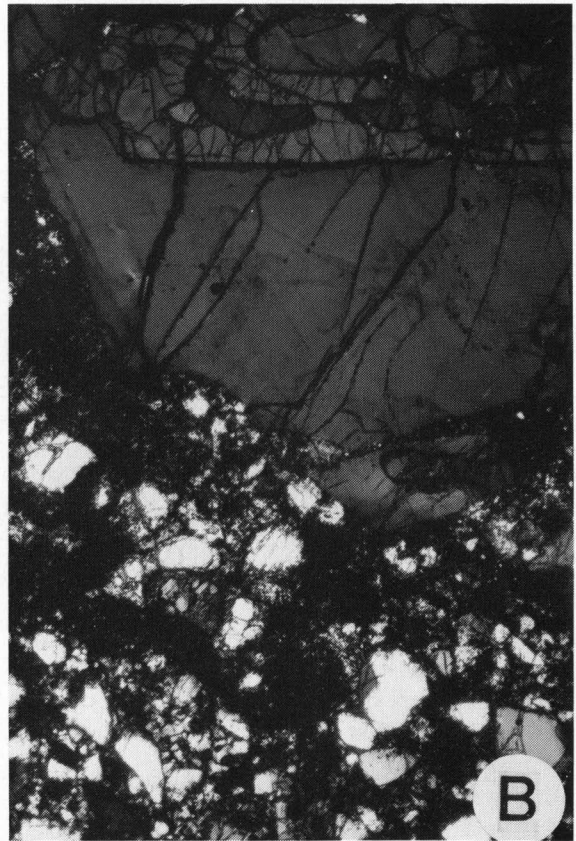


Fig. 1. Photomicrographs of 14321,1027; cross-polarized transmitted light. A. Virtually unbrecciated graphic texture from thin section 14321,1047. Very dark grey = quartz (a single crystal); light grey and white = K-feldspar (two crystals). Field of view: $1.9 \times 1.3 \text{ mm}$. B. Fairly representative area from thin section 14321,993. Relatively unbrecciated graphic-textured area (top): large, darker grey = single crystal of K-feldspar; lighter grey = single crystal of quartz. Thoroughly cataclastic region (bottom): the original texture has been almost completely destroyed; the mineralogy is still nearly pure K-feldspar plus quartz, however. Field of view is $1.7 \times 1.1 \text{ mm}$.

TABLE 1

Compositional data for 14303,204 (specimen 14303,206), 14321,1027 (specimen 14321,1028), and other lunar granitic samples (oxides in wt.%, trace elements in $\mu\text{g/g}$, except Ge and Cd, ng/g; and Re, Ir and Au, pg/g)

	14303,204	14321,1027	12013 light	15405,12	73215c	73255c
Na ₂ O	1.25	0.52	1.4	0.79	0.194	0.53
MgO	3.3	0.07	0.7	1.53	–	0.2
Al ₂ O ₃	18.5	12.5	11.9	10.15	–	12.3
SiO ₂	(58) ^a	74.2	73.0	68.08	–	(75.5) ^a
K ₂ O	3.6	8.6	6.8	3.39	7.0	7.55
CaO	8.8	1.25	1.4	4.89	–	0.5
TiO ₂	0.75	0.33	0.6	0.90	–	0.26
FeO	5.57	2.32	0.9	6.99	2.98	3.1
Mg/(Mg + Fe)	0.51	0.051	0.58	0.28	(0.3) ^b	0.103
Sc	10.7	3.0	21	–	4.8	2.3
V	23	–	59	–	–	7
Cr	550	17	1010	–	103	70
Mn	480	175	940	–	–	300
Co	14.1	0.94	20	–	2.10	1.5
Ni	< 60	4.9	–	–	–	–
Zn	–	1.9	–	–	–	–
Ga	9.1	9.0	–	–	–	–
Ge	–	87	–	–	–	–
Rb	114	210	141	–	–	–
Sr	210	55 ± 16	168	–	–	215
Zr	1020	660	1130	–	–	–
Cd	–	34	–	–	–	–
Cs	2.2	5.7	4	–	–	–
Ba	2100	2160	5780	–	–	5470
La	58	44.3	68	–	42.9	20.3
Ce	149	117	126	–	125	50
Nd	93	58	64	–	–	34
Sm	22	15.9	16.0	–	19.0	6.74
Eu	3.3	1.17	2.54	–	3.11	2.71
Tb	4.9	4.3	4.9	–	5.6	1.52
Dy	32	31.5	29.6	–	–	–
Ho	–	8.4	–	–	–	–
Yb	18	32.2	33.4	–	27.2	10.3
Lu	2.6	5.1	5.1	–	5.3	1.5
Hf	21.6	13.9	24	–	25.6	16.0
Ta	3.3	8.3	6.6	–	5.4	2.4
Re	–	< 18	–	–	–	–
Ir	< 2800	47	–	–	–	–
Au	–	35	–	–	–	–
Th	31	65	41	–	39.9	9.5
U	9.2	23.4	12.3	–	–	–
Reference	new data	new data	see below	[9]	[33]	[42]

References for 12013: oxides from defocused beam electron microprobe analyses of pure felsite [43]; all other elements from averaging data for the light portions (probably *not* pure felsite) of the breccia: [43–46]; only analyses containing at least 3.5% K₂O were averaged.

^a SiO₂ data in parentheses were calculated as the difference between the sum of the other data and 100%.

^b The 73215 felsite Mg/(Mg + Fe) value is an estimate based on mineral composition data [8].

The clast is chemically pristine. Two siderophile elements, Ir and Au (Table 1), are below the concentrations ($3 \times 10^{-4} \times \text{CI}$) recommended [5] as a cut-off for distinguishing most pristine lunar rocks from most nonpristine ones (the latter tend to contain high siderophiles, due to meteoritic contamination). Likewise, Fe-Ni grains are uniformly far poorer in Ni ($< 0.2 \text{ wt.}\%$) than most meteoritic metals ($> 4 \text{ wt.}\%$). The glassy, shock-melted areas are scattered more or less evenly throughout the sections and have compositions appropriate for mixtures of the crystals from the remainder of the clast; they probably are free of foreign contamination. Non-pristine samples tend to have incompatible element patterns dominated by a KREEP component [5], but the pattern for 14321,1027 is far from KREEPlike (Fig. 2). In fact, it is strikingly similar to the patterns of several other lunar granitic samples. Olivine, pyroxene, and ilmenite are uniformly ultra-low in Mg (Fig. 3, Table 2). Hence, if contamination is present, it must have extraordinary mineralogy, just like the granite.

Similarly Mg-poor mafic silicates are common among mare basalts, but none of the lunar granites described previously (nor 14303,204, described below) has such Mg-poor mafics as 14321,1027. In the 15405 granites [9], pyroxenes have $\text{Mg}/(\text{Mg} + \text{Fe}) \sim 0.4$. In the 73215 felsite, high-Ca pyroxene averages En_{23} , and olivine averages Fo_{19} [8]. The

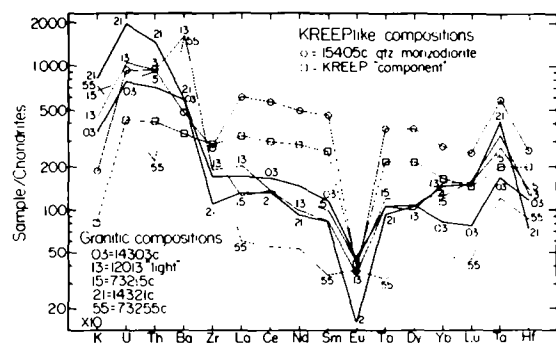


Fig. 2. Incompatible element data for lunar granitic samples, normalized to chondrites. Data sources are listed in Table 1. Also shown are the patterns of the KREEP component as defined by Warren and Wasson [30], and of the KREEPy pristine quartz-monzodiorite clast from 15405 [32].

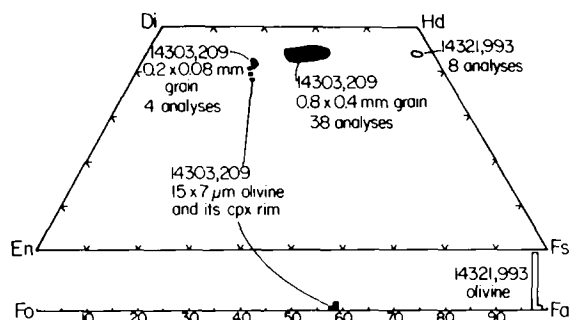


Fig. 3. Compositions (in end member mole %) of pyroxenes and olivines from the 14321,1027 (thin section 993) and 14303,204 (thin section 209) granite clasts.

Apollo 17 Boulder 1 granites contain ferroaugites [10]. The felsite patches in 12013 have pyroxenes with $\text{Mg}/(\text{Mg} + \text{Fe}) \sim 0.45$ [13]. The mafic silicate compositions reinforce our confidence that the extreme composition of 14321,1027 corresponds to a real rock type, not an artifact of unrepresentative sampling.

Feldspar compositions (Fig. 4, Table 2) average $\text{Or}_{88.7}\text{Ab}_{9.3}$. For comparison, K-feldspars from the 73215 felsite [8] and the Apollo 17 Boulder 1 granites [10] are nearly pure barian K-feldspar (Or_{97}). However, the K-feldspars in the 12013 felsite patches are comparatively rich in Na, rang-

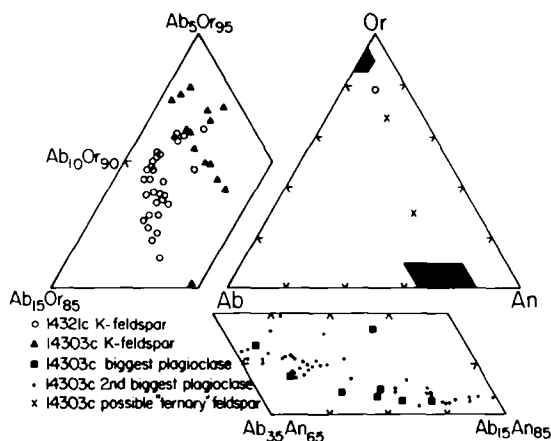


Fig. 4. Compositions of feldspars from the 14321 and 14303 granite clasts. The ternary compositions shown for 14303c are probably artifacts caused by overlap of the electron beam between K-feldspar and plagioclase (see text).

TABLE 2

Compositions of 14321,1027 minerals (obtained from thin section 14321,993)

	Ilmen.	K-spar.	Oliv.	Pyrox.
SiO ₂	–	65.30	29.52	47.59
TiO ₂	50.86	0.05	0.16	0.61
Al ₂ O ₃	0.21	18.66	–	1.44
FeO	47.25	0.13	68.08	29.46
MnO	0.46	–	0.73	0.30
MgO	< 0.1	–	0.83	0.98
CaO	–	0.38	0.24	19.21
Na ₂ O	–	1.01	–	–
K ₂ O	–	14.27	–	–
BaO	–	0.48	–	–
Total	98.78	100.28	99.56	99.59
N	2	30	10	6

^a Abbreviations: Ilmen. = ilmenite, K-spar. = K-feldspar, Oliv. = olivine, Pyrox. = pyroxene, N = number of points averaged.

ing only up to Or₈₀ [13]. Ryder [9] did not report compositions for K-feldspars from the 15405 granites, but we have analyzed the K-feldspars from the 5 × 2 mm clast in 15405,12: they average Or_{93.0}Ab_{5.5}An_{1.5}. Average BaO content of the 14321,1027 K-feldspars is relatively low, only 0.48 wt.% (range among 10 analyses: 0.36–0.60 wt.%).

3. Clast 14303,204

Clast 14303,204 was found as an 11 × 6 mm outcrop on breccia 14303 (see NASA photo S77-23372, reproduced on p. 41 in Twedell et al. [14]). Unfortunately, it is thin, and NASA personnel estimated its original mass to be only 170 mg. For non-destructive (INAA) analysis, we used 70 mg, which, other than having been exposed to 2 × 10¹⁶ neutrons cm⁻², is unaltered and available for further studies.

Exclusive of shock-melted glass (which accounts for roughly half), our 12-mm² thin section (14303,209) contains ~ 40% K-feldspar, 35% quartz, 20% plagioclase (mainly two large crystals), 4% high-Ca pyroxene (mainly one large crystal), 1% ilmenite, plus traces of zircon, Fe-Ni metal, troilite, apatite and olivine. For phases with similar relief and birefringence, i.e., K-feldspar and

quartz, the precise modal abundances are difficult to determine, because the section is less than standard thickness and so severely shocked that large, intact, easily identifiable crystals amount to only about 20% of the area. The norm from the INAA analysis (Table 1) indicates sampling errors. It contains (in vol.%) 47% plagioclase, 23% K-feldspar, 17% pyroxene, 12% quartz, and 1% ilmenite. The best estimate of the overall composition of the clast is obtained by averaging the norm (converted to volume %) and the mode, assuming: (1) that a 70-mg chip and a 12-mm² thin section represent approximately equal fractions of the total clast, and (2) that the average composition of the shock-melted glass is approximately the same as the bulk composition of the remainder of the thin section. This yields the following quasimode: 33% plagioclase, 32% K-feldspar, 23% quartz, 11% pyroxene, and 1% ilmenite. In IUGS classification, this is granite, *sensu stricto*.

Despite generally severe brecciation, small areas display an undamaged igneous texture (Fig. 5). Essentially intact crystals are up to 2 mm across. Quartz occurs in vermicular intergrowths with K-feldspar much like the ones in 14321,1027. The scale of the intergrowths is roughly half that in 14321,1027; i.e., the silica blebs and lamellae are generally 0.05–0.1 mm across. Plagioclase crystals are zoned, in one case (Fig. 5C) from An₈₆ at the core to An₅₉ near the rim. The largest pyroxene crystal (0.8 × 0.4 mm) is zoned from En₂₉Fs₂₉Wo₄₂ to En₂₀Fs₃₆Wo₄₄. Pyroxene contains many vermiform blebs of K-rich glass, mostly ~ 0.03 mm across (Fig. 5D). The origin of this unusual texture is enigmatic. The glassy inclusions are not at all uniform in composition, although it may be that the undevitrified glass per se is uniform, and the variations are due to the presence of tiny crystallites within the inclusions. The glass compositions are generally close to the composition of the bulk clast. Perhaps, therefore, the inclusions formed by occlusion of pockets of the parent magma at the time it crystallized.

Apatite occurs as a single, euhedral 12 × 9 μm crystal. Olivine occurs as a single 15 × 7 μm crystal, rimmed by high-Ca pyroxene. Olivine of this composition (Fo₄₁) is highly unstable with quartz + liquid. It is probably the remnant of a xenocryst,

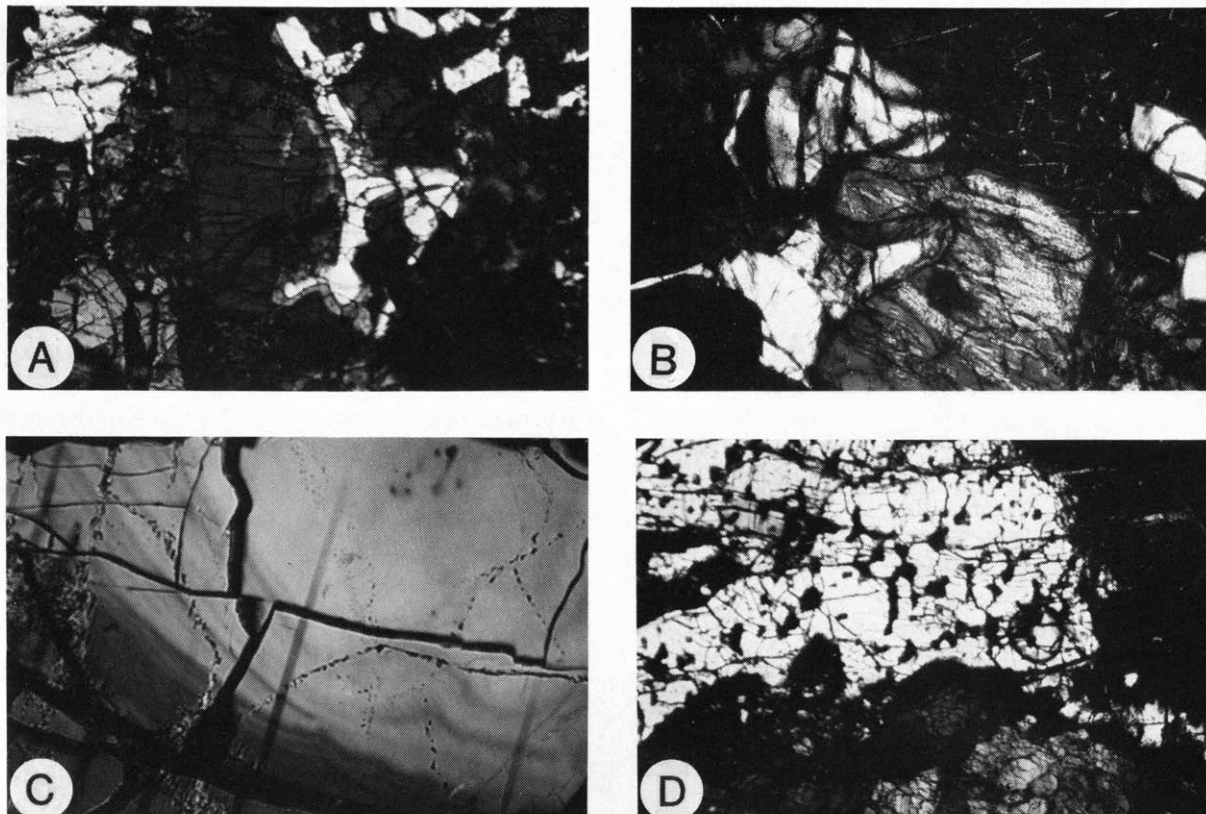


Fig. 5. Photomicrographs of 14303,204; thin section 14303,209; transmitted cross-polarized light. A. Relatively unbrecciated area with graphic texture. White areas are a single crystal of quartz. Most of the remainder is a single crystal of K-feldspar. Field of view is 1.1×0.7 mm. B. Graphic K-feldspar-quartz intergrowth (center) beside a zone of shock-melted glass plus quench crystals (upper right). Field of view is 0.42×0.28 mm. C. Zoned plagioclase crystal (see text). Field of view is 0.53×0.35 mm. D. Pyroxenes, the largest of which is shown here (white), are full of pockets of partially devitrified glass, giving them a "worm-eaten" appearance. Dark area (upper right) consists of shock-melted glass plus quench crystals. Lighter grey area (along bottom) is part of brecciated plagioclase that is 2 mm across. Field of view is 1.1×0.7 mm.

either assimilated but incompletely dissolved by the granitic magma as it was emplaced in the crust, or else introduced by the impact that produced the glass. However, G. Ryder (review of this paper) argues that it is "more likely [a product of] disequilibrium crystallization in a viscous liquid". The olivine is situated along an edge of the thin section; even if this one area is polymict, the evidence cited below shows that the bulk of the clast is pristine.

The best evidence for pristinity is textural: even though the clast as a whole is severely brecciated, areas with obviously pristine graphic intergrowth textures are scattered throughout the thin section.

To conserve material for other studies, two destructive techniques we normally use (fused bead electron microprobe analysis and RNAA) were not employed. Unfortunately, the non-destructive (INAA) technique is not sensitive enough to determine siderophile elements at levels typical for pristine lunar rocks. However, INAA has provided upper limits for some siderophile elements which are consistent with pristinity. The Fe-Ni grains are diverse in composition, but consistently lower in Ni (average = 0.92, maximum = 2.8 wt.%) than most metal of meteoritic origin (> 4 wt.%). Compared to other lunar granitic samples, 14303,204 has a REE pattern (Fig. 2) similar to

TABLE 3

Compositions of 14303,204 minerals

	Apat.	Ilmen.	K-spar.	Plag. extremes		Pyrox. extremes ^a		Zirc.
				An ₈₃	An ₅₉	En ₃₇	En ₂₀	
SiO ₂	1.33	—	64.23	47.51	53.80	51.37	49.29	33.52
TiO ₂	0.08	54.19	0.12	—	—	1.45	0.77	0.08
Al ₂ O ₃	0.11	0.08	19.48	33.77	28.63	1.51	0.58	—
Cr ₂ O ₃	—	—	—	—	—	0.32	0.15	—
FeO	0.61	44.13	—	0.12	0.22	12.48	20.89	0.32
MnO	—	—	—	—	—	0.24	0.31	—
MgO	—	1.83	—	—	—	12.04	6.68	—
CaO	51.48	—	0.32	17.53	12.05	19.16	20.47	—
Na ₂ O	—	—	0.68	1.76	3.86	0.17	0.09	—
K ₂ O	0.32	—	13.66	0.31	1.06	—	—	—
BaO	—	—	3.27	—	—	—	—	—
P ₂ O ₅	39.13	—	—	—	—	—	—	—
ZrO ₂	—	—	—	—	—	—	—	65.75
Total	98.04 ^b	100.23	101.76	101.00	99.62	98.74	99.23	99.67
<i>N</i>	6	6	13	1	1	1	1	3

Abbreviations: Apat. = apatite, Ilmen. = ilmenite, K-spar. = K-feldspar, Plag. = plagioclase, Pyrox. = pyroxene, Zirc. = zircon, *N* = number of points averaged.

^a Pyroxene from the reaction rim around the olivine xenocryst not included.

^b Includes 3.5% F, 0.3% Cl, 1.0% Y₂O₃, 0.20% La₂O₃, 0.18% Ce₂O₃, 0.07% Nd₂O₃, and 0.03% Sm₂O₃.

that of KREEP. The La/Sm ratio is 17% higher, but the La/Lu ratio is essentially identical to KREEP's. However, the clast has exceptionally high Th, U and Ta relative to the REE. Its mineralogy and overall composition are unlikely to be products of mixing of common lunar materials. Compositions of the shock-melted glasses are diverse, but always consistent with a mixture of the intact minerals in their midst.

Mafic silicate compositions are shown in Fig. 3 and Table 3. The zoning in the 0.8 × 0.4 mm pyroxene is roughly concentric, but oscillatory. As discussed above, the single tiny olivine crystal may be a xenocryst. However, even the large high-Ca pyroxenes are far more magnesian than their 14321,1027 counterparts. The near-overlap between the compositions of the large pyroxenes and the pyroxene reaction rim around the olivine could be a coincidence. We doubt that the larger pyroxenes are xenocrysts: (a) they contain what appear to be melt inclusions that are granitic in

composition and, hence, presumably formed in the presence of a granitic magma; (b) the ilmenite is commensurably magnesian; and (c) pyroxenes from most other lunar granites are comparably magnesian.

K-feldspar major element compositions (Fig. 4, Table 3) are similar to their 14321,1027 counterparts, averaging Or_{90.7}Ab_{7.1}. However, BaO contents are relatively high (average = 3.2 wt.%) and diverse (range among 12 analyses: 1.5–4.8 wt.%). Near, though not quite on, the rim of the most intensely zoned plagioclase crystal is an ~ 25 × 5 μm zone which contains K-feldspar, plus possibly some "ternary" feldspar. Ternary feldspars occur in the granites from Apollo 17 Boulder 1 [10] and 14305 [12]. However, the suspected ternary patch in 14303 is extremely heterogeneous at the scale of ~ 1 μm, and the apparent ternary compositions are most likely artifacts caused by overlap of the electron beam between minute lamellae of K-feldspar and plagioclase.

4. Petrogenesis

4.1. Where and when

Before brecciation, both clasts were medium-grained (dominantly 1–2 mm; coarse, by lunar standards), and most, if not all, K-feldspar and quartz occurred in symplectitic intergrowths, borderline between graphic *sensu stricto* and micrographic. Graphic intergrowths occur as megacrysts in terrestrial volcanic rocks [15], but they are much more common among intrusives. The strong zoning in the plagioclase and pyroxene, plus the glassy inclusions in the pyroxene, are possible indications that 14303,204 cooled rapidly. But Ca-feldspars of plutonic acidic igneous rocks from Earth are often zoned, too [16]. Considering the high viscosities of their parent magmas, their grain sizes, and particularly the coarseness of their graphic intergrowths, both lithologies are probably of intrusive origin.

It is probably no coincidence that both of these “large” pristine lunar granites are from Apollo 14. Remote sensing data [17] show that on the Moon’s surface K, Th and U are concentrated in the longitude range 0–50 deg. W. Only Apollo 14 (18 deg. W) and 12 (23 deg. W) returned samples from within this range. The only truly large (82 g) granitic sample, non-pristine breccia 12013, is among the few non-mare samples from Apollo 12. About 4% of glasses from Apollo 14 soils are granitic in composition (not all are shock-melted granites, however) [18].

It was once assumed that the lunar crust, except for comparatively young KREEP and mare lithologies, formed entirely out of a primordial magma ocean. This is now considered to be an oversimplification [19–21]. The Mg-rich suite of pristine rocks probably formed in later, less massive intrusions, analogous to terrestrial layered complexes. Some layered basic intrusions contain acidic final differentiates, e.g., the Dufek [22], Fongen-Hyllingen [23], Skaergaard [24], and Dore Lake [25] intrusions. Thus, some lunar granites may have formed as the last differentiates of the magma ocean, but some might also have formed from late-stage magmas in younger, smaller intrusions, produced from partial melts of the deep interior

(and, possibly, assimilated magma ocean differentiates). Age measurements for 14321,1027 [26] are 4.09 ± 0.11 Ga, and 4.11 ± 0.20 Ga, by internal isochrons from Rb-Sr and Sm-Nd, respectively; and the Rb-Sr model age is 4.11 ± 0.03 Ga. Unfortunately, the other dated granitic samples all may have had their isotopic records disturbed by intense brecciation [8]. The least disturbed dated sample is probably the 73215 felsite, for which the most conservatively high Rb-Sr model age, 4.05 Ga [27], is interpreted as a probable upper limit on the crystallization age [8]. Thus, these two granites appear too young to be direct products of the primordial magma ocean, unless it took about 400 Ma to crystallize. The last dregs of the magma ocean may have remained molten this long [28], but after ~200 Ma there was probably more partial melt (from the deep interior) than ocean residuum (fig. 8 in Solomon and Longhi [28]); the former was liable [19] to intrude the crust (producing Mg-rich layered complexes), and in so doing, assimilate the residuum. The marked cluster of model ages of KREEPy samples at ~4.36 Ga is widely interpreted as the end of the magma ocean era, *per se* (e.g. [29–31]).

Besides relatively young ages, models invoking multiple post-ocean intrusions imply diversity. The considerable contrast in mineralogy among 14303,204, 14321,1027, and several other lunar granites has been described above. Bulk compositions (Fig. 2, Table 1) are remarkably similar among analyzed granites and felsites, except for 14303,204, which resembles KREEP (note especially the heavy REE). KREEP is believed to be a derivative of the final residuum (“urKREEP”) from the magma ocean [30]. Thus, 14303,204, far more than any of the others, has a composition that suggests kinship with the residuum from the magma ocean. On balance, however, available evidence indicates that the magma ocean did not *directly* produce *all* the granites.

4.2. Liquid immiscibility?

As reviewed by Taylor et al. [32], much has been written about the possibility that silicate liquid immiscibility (SLI) was important in the genesis of evolved lunar rocks, such as the granites.

Taylor et al. found no strong evidence for extensive, large scale SLI. Neither 14321,1027 nor 14303,204 have textural features such as ocelli or globules that would be indicative of SLI, but the samples are too small for this to be a major objection. Like other lunar granites (Table 1), these two have extraordinarily high K/REE and K/P ratios (P contents are inferred to be low on the basis of modes), as expected from SLI [26,33]. However, two other aspects of the bulk compositions of these and all the other analyzed lunar granites (Table 1, Fig. 2) suggest that crystal-liquid fractionation was more important than SLI in their genesis. First, they all have extraordinarily high U,Th/REE ratios, which is the opposite of what is expected for Si-rich SLI products on the basis of cation charge densities [34,35]. Second, except for 14303,204, they all have extraordinarily V-shaped REE patterns. This is not an expected outcome from SLI, but it is from fractionation of apatite (distribution coefficients: [36–38]). The scarcity of apatite in the granites is an embarrassment for this model, but even our “large” samples are sufficiently small to permit gross sampling errors among minor phases. Unless the analyses are unrepresentative, they imply a paradox that is difficult to explain without SLI: Despite the comparatively evolved nature of the granites (higher Si, lower Mg/Fe, etc.), none of them is as rich in any but the heaviest REE as the KREEP “component” of Warren and Wasson [30] (Fig. 2); in fact, one pristine KREEPy sample (15405c) is far richer in *all* REE.

Incidentally, most KREEPy rocks are fine-grained polymict breccias, which may occasionally include cryptic granite components. In such a case, the incompatible element pattern would have, in comparison to the patterns of granite-free KREEP: (a) higher U, Th and Ba; (b) a less negative slope among heavy REE; and (c) barring losses by volatilization, higher alkalis (K, Rb and Cs). Therefore, emphasis should go to pristine samples, in interpreting the origin of KREEP.

4.3. Origin of graphic textures

On Earth, rocks with graphic textures are almost invariably rich in phases such as amphiboles

or micas that contain H₂O plus other volatiles. Many hypotheses for the origin of the intergrowths call for eutectic crystallization, or crystallization from a late-stage supercritical volatile-rich fluid. Others (e.g. [39]) call for metasomatic replacement, with transport of Si, Al and K being accomplished by aqueous vapors. Barker [15] emphasized that all graphic intergrowths are probably not formed by just one mechanism. Judging from the lunar granites, volatiles are scarcely a prerequisite for formation of the texture, because every reasonably large lunar granite yet described has graphic intergrowths, even though volatile-rich phases are absent. Among the lunar granites, the graphic texture seems to be a product of simultaneous crystallization from a melt of K-feldspar + quartz.

4.4. Implications for the lunar H₂O budget

The absence of hydrous phases and paucity of volatile elements in general was one of the first big surprises to emerge from study of the Apollo-11 basalts—“the driest rocks known to man” [40]. Virtually all samples from the other eight landing sites are also devoid of hydrous phases. However, the notion of H₂O and CO₂ being important in lunar evolution is not quite dead. Smith [41] speculated that hydrous phases may have once been more abundant, their present scarcity being due to losses of volatiles via shock and thermal metamorphism, during the heavy meteoritic bombardment that took place shortly after the lunar crust formed. Moreover, the lunar samples are preponderantly of lithologic types which even on Earth are not particularly H₂O-rich: basalts and anorthositic-noritic-troctolitic cumulates. If we wanted to sample hydrous phases among unmetasomatized terrestrial igneous rocks, we would look not for basalts or mafic cumulates, but for acid intrusives, particularly pegmatites. The absence of hydrous phases even among pristine lunar granites with textures that are “micropegmatitic,” and in the case of 14321,1027 arguably even graphic (*sensu stricto*), suggests that the limit on the lunar content of H₂O may be even lower than previously supposed.

Acknowledgements

We thank the JSC curatorial staff and LAPST for their assistance; J. Longhi and (as usual) G. Ryder for helpful reviews; and J.N. Grossman, G.W. Kallemeyn, T. Ko, F. Kyte, D. Malvin, E. Rambaldi and A. Zane of UCLA, and G. Conrad and J. Allen of UNM, for technical support. This research was supported by NASA grants NGL 32-004-063 (K. Keil) and NGL 05-007-367 (J. Wasson).

References

- 1 S. Miyamoto, A geological interpretation of the lunar surface, *Planet. Space Sci.* 2, 256–260, 1960.
- 2 R.G. Brereton, The lunar surface, *Geotimes* 6, 22–26, 1961.
- 3 J.A. O'Keefe and W.S. Cameron, Evidence from the Moon's surface features for the production of lunar granites, *Icarus* 1, 271–285, 1962.
- 4 D.R. Chapman and H.K. Larson, On the lunar origin of tektites, *J. Geophys. Res.* 68, 4305–4353, 1963.
- 5 P.H. Warren and J.T. Wasson, Pristine nonmare rocks and the nature of the lunar crust, *Proc. 8th Lunar Sci. Conf.*, pp. 2215–2235, 1977.
- 6 P.H. Warren, G.J. Taylor, K. Keil, G.W. Kallemeyn, P.S. Rosener and J.T. Wasson, Sixth foray for pristine nonmare rocks, and an assessment of the diversity of lunar anorthosites, *Proc. 13th Lunar Planet. Sci. Conf.* (in press).
- 7 A.L. Streckeisen, Plutonic rocks: classification and nomenclature recommended by the IUGS Subcommittee on the Systematics of Igneous Rocks, *Geotimes* 18/10, 26–30, 1973.
- 8 O.B. James and J.G. Hammarstrom, Petrology of four clasts from consortium breccia 73215, *Proc. 8th Lunar Sci. Conf.*, pp. 2459–2494, 1977.
- 9 G. Ryder, Lunar sample 15405: remnant of a KREEP basalt-granite differentiated pluton, *Earth Planet. Sci. Lett.* 29, 255–268, 1976.
- 10 G. Ryder, D.B. Stoeser, U.B. Marvin and J.F. Bower, Lunar granites with unique ternary feldspars, *Proc. 6th Lunar Sci. Conf.*, pp. 435–449, 1975.
- 11 D.B. Stoeser, R.W. Wolfe, J.A. Wood and J.F. Bower, Petrology, in: *Interdisciplinary Studies of Samples from Boulder 1, Station 2, Apollo 17*, Vol. 1, pp. 35–109, J.A. Wood, ed., Smithsonian Astrophysical Observatory, Cambridge, Mass., 1974.
- 12 J.W. Shervais and L.A. Taylor, Micrographic granite: More from Apollo 14, in: *Lunar and Planetary Science XIV*, pp. 696–697, Lunar and Planetary Institute, Houston, Texas, 1983.
- 13 J.E. Quick, O.B. James and A.L. Albee, Petrology and petrogenesis of lunar breccia 12013, *Proc. 12th Lunar Planet. Sci. Conf.*, pp. 117–172, 1981.
- 14 D. Twedell, S. Feicht, I. Carlson and C. Meyer, Jr., Lithological maps of selected Apollo 14 breccia samples, JSC Publ. No. 13842, NASA Johnson Space Center, Houston, Texas, 1978.
- 15 D.S. Barker, Compositions of granophyre, mymerkite, and graphic granite, *Geol. Soc. Am. Bull.* 81, 3339–3350, 1970.
- 16 S.R. Nockolds, R.W.O'B. Knox and G.A. Chinner, *Petrology for Students*, Cambridge University Press, London, 1978.
- 17 J.R. Arnold, A.E. Metzger and R.C. Reedy, Computer-generated maps of lunar composition from gamma-ray data, *Proc. 8th Lunar Sci. Conf.*, pp. 945–948, 1977.
- 18 S.B. Simon and J.J. Papike, The Apollo 14 regolith: petrology of cores 14210/14211 and 14220 and soils 14141, 14148, and 14149, *J. Geophys. Res.* 87 (Suppl.), A232–A246, 1982.
- 19 P.H. Warren and J.T. Wasson, Early lunar petrogenesis, oceanic and extraoceanic, *Proc. Conf. Lunar Highlands Crust*, pp. 81–99, 1980.
- 20 O.B. James, Rocks of the early lunar crust, *Proc. 11th Lunar Planet. Sci. Conf.*, pp. 365–393, 1980.
- 21 P.H. Warren, G.J. Taylor, K. Keil, C. Marshall and J.T. Wasson, Foraging westward for pristine nonmare rocks: complications for petrogenetic models, *Proc. 12th Lunar Planet. Sci. Conf.*, pp. 21–40, 1981.
- 22 A.B. Ford, Development of the layered series and capping granophyre of the Dufek intrusion of Antarctica. In: *Symposium on the Bushveld Igneous Complex and Other Layered Intrusions*, D.J.L. Visser and G. Von Gruenewaldt, eds. *Geol. Soc. S. Afr. Spec. Publ.* 1, 492–510, 1970.
- 23 J.R. Wilson, K.H. Esbensen and P. Thy, Igneous petrology of the synorogenic Fongen-Hyllingen layered basic complex, south-central Scandinavian Caledonides, *J. Petrol.* 22, 584–627, 1981.
- 24 L.R. Wager and G.M. Brown, *Layered Igneous Rocks*, W.H. Freeman and Co., San Francisco, Calif., 1967.
- 25 G.O. Allard, The Dore Lake complex, Chibougamau, Quebec—a metamorphosed Bushveld-type layered intrusion, in: *Symposium on the Bushveld Igneous Complex and Other Layered Intrusions*, D.J.L. Visser and G. Von Gruenewaldt, eds., *Geol. Soc. S. Afr. Spec. Publ.* 1, 477–491, 1970.
- 26 L. Nyquist, C.-Y. Shih, B. Bansal, H. Weismann and J. Wooden, Formation of a lunar granite 4.1 AE ago, in: *Lunar and Planetary Science XIV*, pp. 576–577, Lunar and Planetary Institute, Houston, Texas, 1983.
- 27 W. Compston, J.J. Foster and C.M. Gray, Rb-Sr systematics in clasts and aphanites from consortium breccia 73215, *Proc. 8th Lunar Sci. Conf.*, pp. 2525–2549, 1977.
- 28 S.C. Solomon and J. Longhi, Magma oceanography, 1. Thermal evolution, *Proc. 8th Lunar Sci. Conf.*, pp. 583–599, 1977.
- 29 F. Herbert, M.J. Drake, C.P. Sonett and M.J. Wiskerchen, Some constraints on the thermal history of the lunar magma ocean, *Proc. 8th Lunar Sci. Conf.*, pp. 573–582, 1977.
- 30 P.H. Warren and J.T. Wasson, The origin of KREEP, *Rev. Geophys. Space Phys.* 17, 73–88, 1979.

- 31 R.W. Carlson and G.W. Lugmair, Sm-Nd constraints on the early lunar differentiation and the evolution of KREEP, *Earth Planet. Sci. Lett.* 45, 123–132, 1979.
- 32 G.J. Taylor, R.D. Warner, K. Keil, M.-S. Ma and R.A. Schmitt, Silicate liquid immiscibility, evolved lunar rocks and the formation of KREEP, *Proc. Conf. Lunar Highlands Crust*, pp. 339–352, 1980.
- 33 D.P. Blanchard, J.W. Jacobs and J.C. Brannon, Chemistry of ANT-suite and feldspar clasts from the consortium breccia 73215 and of gabbroic anorthosite 79215, *Proc. 8th Lunar Sci. Conf.*, pp. 2507–2524, 1977.
- 34 P.C. Hess and M.J. Rutherford, Element fractionation between immiscible melts, in: *Lunar Science V*, pp. 328–330, Lunar Science Institute, Houston, Texas, 1974.
- 35 E.B. Watson, Two-liquid partition coefficients: experimental data and geochemical implications, *Contrib. Mineral. Petrol.* 56, 119–134, 1976.
- 36 H. Nagasawa, Rare earth concentrations in zircons and apatites and their host dacites and granites, *Earth Planet. Sci. Lett.* 9, 359–364, 1970.
- 37 A.J. Irving, A review of experimental studies of crystal/liquid trace element partitioning, *Geochim. Cosmochim. Acta* 42, 743–770, 1978.
- 38 V.I. Kovalenko, V.S. Antipin, N.V. Vladykin, E.V. Smirnova and Yu.A. Balashov, REE partition ratios in apatites and REE behavior in magmatic processes, *Geokhimiya*, pp. 230–243, 1982.
- 39 R.H. Jahns, R.F. Martin and O.F. Tuttle, Origin of granophyre in dikes and sills of tholeiitic diabase, *Trans. Am. Geophys. Union* 50, 337, 1969 (abstract).
- 40 E. Roedder, Silicate liquid immiscibility in magmas and in the system K_2O -FeO- Al_2O_3 - SiO_2 : an example of serendipity, *Geochim. Cosmochim. Acta* 42, 1597–1617, 1978.
- 41 J.V. Smith, Was a hydrous and carbonated magma present on the Moon?, in: *Lunar and Planetary Science XII*, pp. 1005–1007, Lunar and Planetary Institute, Houston, Texas, 1981.
- 42 D.P. Blanchard and J.R. Budahn, Remnants from the ancient lunar crust: clasts from consortium breccia 73255, *Proc. 10th Lunar Planet. Sci. Conf.*, pp. 803–816, 1979.
- 43 J.E. Quick, A.L. Albee, M.-S. Ma, A.V. Murali and R.A. Schmitt, Chemical compositions and possible immiscibility of two silicate melts in 12013, *Proc. 8th Lunar Planet. Sci. Conf.*, pp. 2153–2189, 1977.
- 44 N.J. Hubbard, P.W. Gast and H. Wiesmann, Rare earth, alkaline and alkali metal and $87/86$ Sr data for sub-samples of lunar sample 12013, *Earth Planet. Sci. Lett.* 9, 181–184, 1970.
- 45 C.C. Schnetzler, J.A. Philpotts and M.L. Bottino, Li, K, Rb, Sr, Ba and rare-earth concentrations, and Rb-Sr age of lunar rock 12013, *Earth Planet. Sci. Lett.* 9, 185–192, 1970.
- 46 H. Wakita and R.A. Schmitt, Elemental abundances in seven fragments from lunar rock 12013, *Earth Planet. Sci. Lett.* 9, 169–176, 1970.